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$\boldsymbol{\sigma}$-álgebra

Definición 1 ($\boldsymbol{\sigma}$-álgebra). Sea $X \neq \varnothing$, $\Sigma \subset \mathcal{P}(X)$ es una σ-álgebra de $X \iff$

(i) $X \in \Sigma$

(ii) $E \in \Sigma \implies E^{\mathsf{c}} = X \setminus E \in \Sigma$

(iii) $\left\{E_n\right\}_{n=1}^{\infty} \subset \Sigma \implies \displaystyle\bigcup_{n=1}^{\infty} E_n \in \Sigma$

Ejemplos 1 (de $\boldsymbol{\sigma}$-álgebras). Sea X un conjunto arbitrario.

$\boxed{1}$ $\mathcal{P}(X)$ es una σ-álgebra de X.

$\boxed{2}$ $\Sigma = \{\varnothing, X\}$ es una σ-álgebra de X.

$\boxed{3}$ $\Sigma = \{\varnothing, E, E^{\mathsf{c}}, X\}$ es una σ-álgebra de X con $E \subset X$.

Sea $X = [0,1) \subset \mathbb{R} \implies \Sigma = \left\{\varnothing, \left[0, \frac{1}{2}\right), \left[\frac{1}{2}, 1\right), [0,1)\right\}$ es una σ-álgebra de X.

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